COM/2023/769: Directive on the "welfare of dogs and cats and their traceability"
- Article 15: Painful Practices - Electric Current

Submission from the Electronic Collar Manufacturers Association™, 745 rue de la Bergeresse, 45160 Olivet, France

Executive Summary

This submission on the directive on the "welfare of dogs and cats and their traceability" looks at the current text of Article 15.3 (c) on "Painful Practices". It proposes to prohibit "exposing dogs and cats to an electric current". [1] We respectfully suggest that this text should be removed for several reasons.

Our initial concern is that the text would ban the use of e-collar training in many breeding and boarding facilities. However, the justification for the ban does not distinguish between their use in these facilities and elsewhere. Campaigners are therefore already suggesting that the specific prohibition would pave the way for a complete ban.

Furthermore, the text is so broadly drafted that, as well as e-collars, its scope includes electric fencing used to protect pets and vet-recommended therapeutic treatments which also use electric current on animals. This electrotherapy is used post-operatively to help maintain muscle mass and to alleviate chronic conditions such as osteoarthritis. [2] There is therefore a need for the institutions to review both the intent and consequences of the current text.

As the Electronic Collar Manufacturers Association™, we view this proposal as an issue of principle as well as an immediate risk to the businesses of our members across the European Union.

In this document we give detailed evidence as to why any ban on e-collars would be severely counterproductive for animal welfare and why the legislation is opposed by pet owners, hunters and farmers across the member states. We will show that several million EU citizens use these devices.

They, along with many scientists and vets, are convinced that e-collar training offers a uniquely effective way to safeguard both the lives of their pets and those creatures which they would otherwise attack. Last month opposition by a coalition of these groups led to the UK government halting legislation for a ban in England. [3]

This submission seeks to help the Commission understand the two main areas of the international debate on e-collars:

1) the effectiveness of e-collar training: here we focus on relevant behavioural psychology along with scientific research evaluating how e-collar systems prevent predatory attacks on livestock. We also show real world data comparing those areas when bans have been imposed with neighbouring areas where they are permitted.

2) whether e-collar training is cruel: here we look at the legal definition of cruelty. We then review how the aversive nature of e-collars compares with a) other forms of legal use of electric current on animals and b) the aversive interventions of vets.

The submission also looks at the alternatives to e-collar training.

Introduction

E-collar systems are used to train and contain animals in order to keep them safe and deter them from harming other animals.

Around the world, tens of millions of pet owners use them to stop predatory attacks by their dogs and safely contain their cats. Research has also shown them to be effective as a non-lethal method of preventing attacks on livestock by wolves and coyote. [4]

In addition, there is growing use of geo-fenced e-collar systems to manage livestock grazing by farmers and NGOs. [5] The domestic systems are used by pet owners to keep their animals within gardens.

The training versions are very flexible. In New Zealand, the Department of Conservation mandates their use to deter dogs from killing kiwi. [6] In Australia, vets use them to make dogs wary of attacking snakes. [7]

Every region has different needs. In the EU, it is likely that some of the 30,000 livestock deaths attributed to wolves each year are the result of dog attacks: there are regular fatal attacks by dogs in areas where there are no wolves. [8]

The dogs behind these predatory attacks often end up shot dead by farmers or put down by vets. These are perfectly healthy dogs dying for pursuing their natural predatory targets. The problem is not limited to large canines. Pursuits by small domestic dogs cause terrified ewes to abort and suffocate in corners of fields.

An academic study found that 3.3% of dog owners use them to train dogs, 0.9% as containment fence systems and 1.4% as anti-bark collars. [9] In total 4.8% of dogs had an e-collar for one of more purpose.

Given the EU has 72 million dogs this would indicate around 3.6 million dogs are being trained with e-collars across the member states. [10]

There is no data available on how widely containment systems are used for keeping cats in gardens. The best-known user is Andrew Lloyd Webber, the composer of the musical “Cats”. [11]

For pet owners the main uses of this technology are:

i) as training devices for dogs operated by the owner. These remote-controlled devices deliver a pulse through a small battery in the collar to prevent predatory behaviour by their dog. This makes the dogs averse to engaging in future predatory attacks even if it has escaped its owner. This is critical, as most attacks on livestock occur when the owner is not present. This aversion training protects the lives of both the target animals and the dogs which otherwise often get shot by farmers or put down by vets.

ii) as part of a containment fence system to keep cats and dogs from being run over or attacking other animals. The pet receives a warning sound through its collar if it approaches a pre-determined garden perimeter. If the pet goes right up to the perimeter it gets ‘a startle’ through its collar.

iii) to stop dogs from barking.

The challenge for officials in regulating this area comes not just from its scale and complexity but also the intense commitment of owners to using the devices. They see their pets as part of their families and use e-collar technology because they believe that there is no other way to keep their animals safe.

For example, they may regard alternatives such as physical fencing to be prohibitively expensive or consider that keeping a dog permanently on a lead is cruel because running free is vital to its physical and mental health.

Effectiveness

Effective training of dogs has never been more vital given the rapid rise in pet ownership during the pandemic. The resulting combination of growing puppies and inexperienced owners has created a crisis for farmers.

a) Behavioural Psychology

E-collar training draws on the widely accepted theories of "operant conditioning" and "classical conditioning".

Operant conditioning concerns voluntary responses to consequences. It helps explain how, in theory, training gets the dog to associate approaching a sheep with an uncomfortable electronic pulse. In the future, the dog would then choose to refrain from going near sheep to avoid the pulse which it considers the sheep produce.

This voluntary response from the dog is combined with an involuntary one through the process known as classical conditioning. Under this theory (made famous by the example of Pavlov's Dog) the sheep take on the characteristics of the unwanted pulse. Therefore the dog, which used to instinctively pursue sheep, now instinctively avoids them.

The claim made by advocates of e-collar training is that the dog will develop this lasting aversion to approaching sheep after just one or two pulses.

b) The Science

There is a substantial body of research looking at the effectiveness of e-collar training in stopping predatory attacks. [12]

- "All 1,156 dogs displayed avoidance… after the first training session", *Dale*
- "The collar averted all 13 attempted attacks on lambs", *Andelt*
- e-collars "resulted in complete and permanent elimination of aggression", *Tortora*
- "no dogs showed interest in… sheep in the path test", *Christiansen*
- the aversion response "lasts at least 1 year after training", *Dale and Statham*
- e-collar training "shows stronger learning effect", *Salgirli*
- "the most effective" training, *Howell and Bennett*

The effectiveness of the training is not just limited to domestic dogs. Research has found that they are effective with reducing predation by coyote and grey wolves. This is less of a surprise given that wolves and dogs share 99.96% of their DNA. [13]

c) The Real World Evidence

Although not currently in the EU, the UK provides the best comparative data to observe the real-world impact of e-collar training. Wales prohibited the use of e-collars in 2010 but it remains legal elsewhere in the UK. Two years ago the UK media reported that Wales was suffering from four times more dog attacks and had three times more dogs being shot than comparable parts of the UK. [14]

This led to the former Welsh Secretary, David Jones MP, saying that Cardiff's ban "has failed and, paradoxically, is leading to many more animal deaths." [15] His Plaid Cymru counterpart Peredur Owen Griffiths MS observed that: "e-collars act like a vaccine against a dog's desire to chase sheep… being shot by a farmer is clearly far worse for the dog than a one-off startle from an e-collar." [16]

Fresh data released in 2024 shows a further surge of attacks in Wales. It means that over the last three years, while attacks on livestock in the rest of the UK were up by a third, in Wales there was a four-fold increase in dog attacks. [17]

NFU Mutual on dog attacks	**2020**	**2021**	**2022**	**2023**	**Percentage Increase 2020-2023**
UK Total	£1,300,000	£1,520,000	£1,800,000	£2,400,000	Up 85%
Wales	**£165,000**	£306,000	£439,000	**£883,000**	**Up 335%**
UK exc. Wales	**£1,135,000**	£1,214,000	£1,361,000	**£1,517,000**	**Up 34%**

Cruelty

a) The legal concept of cruelty

E-collars are part of a wide range of training and restraint methods used upon animals which the animal would not voluntarily choose to submit to. Other aversive methods include the use of leads and the animal being shouted at by its owner. The legal question is whether the discomfort caused to the animal is in its interests and proportionate.

Statutes such as the Republic of Ireland's Animal Health and Welfare Act 2013 define cruelty as being when the discomfort constitutes "unnecessary suffering". [18] Without such thresholds any discomfort could be defined as cruelty even though it was manifestly in the welfare interests of the animal and/or other animals.

Therefore, one side of the equation in deciding whether e-collars are cruel is to look at the animal welfare impact of dog attacks. These can kill 80 sheep at a time. [19] Ewes that survive attacks often abort and whole flocks can be traumatised. In addition, the dog is typically shot or euthanised.

Bearing in mind the scientific evidence pointing to the efficacy of e-collar training, there would need to be extraordinary harm to the dog from wearing an e-collar to warrant banning them.

Such harm would already be an offence under existing animal welfare law. Yet, despite the widespread use of these systems, we see no evidence of the courts finding e-collars to be cruel. Furthermore, the authorities in Scotland and England have never found cause to prosecute for harm caused by e-collars. [20] [21]

Similarly, organisations campaigning for bans have not been able to provide real-world evidence of harm. For example, the Kennel Club admitted to the Scottish Government that it could provide no evidence of misuse or abuse. [22]

So in assessing where the balance of interest rests for animal welfare, judges are not finding e-collar training to be cruel. But what does scientific research say?

b) The peer-reviewed science

We have seen that the research finds e-collar training remarkably effective at preventing serious harm to animals. But what does it say about the degree of harm caused to the pet by the use of the e-collar: [23]

- “no negative effect on the dogs”, *Christiansen*
- Dogs "did not show considerable or persistent stress indicators”, *Schalke*
- “induced less stress” than other training methods, *Salgirli*
- e-collar use led to “significant increases in the dogs' emotional stability”, *Tortora*

The UK Companion Animal Welfare Council says the word “shock” is “misleading” because the pulse produced is not strong enough. [24] This is why academics increasingly refrain from calling these modern devices “shock collars”. We note that officials in London use the impartial term “electronic collars” in official documents [25] while their counterparts in Edinburgh call the technology “e-collars”. [26]

c) The electric comparables

Modern e-collars do not produce a ‘shock’. The international product standard limits the maximum pulse from an e-collar to 5 mJ (millijoules). [27] To provide context, livestock fences sold in the EU - which any animal can accidentally touch - go up to 15,000 mJ. [28] That is three thousand times higher. And it is a shock.

Today’s e-collars are not just carefully calibrated in terms of power, but the pulse is only delivered between two nodes 60 mm apart. This means that there is no full body ‘shock’ to the animal. Even at the highest setting, the dog will only feel an uncomfortable static click in a small part of its neck.

The campaigners against e-collars ignore the peer-reviewed science and the technical standards. Instead, try to make their case through emotive language. One MEP has gone as far as describing e-collars as “torture devices”. [29]

This may be because some of them fail to understand how electricity works. For instance, speaking in Ireland, Senator Regina Doherty says that e-collars were “cruel” because they create “6,000 volts”. [30]

However, electric pulses are not measured in volts. The term refers to potential energy. On their own, volts cannot be felt. That also needs a flow of current which is measured in amps. It is the combination of volts and amps that delivers energy, the amount of which is measured in joules.

This is why in school science lessons children can safely touch a Van de Graaff generator delivering not only 6,000 volts but 200,000 volts. [31] The device raises the child’s hair but does not harm them. Similarly, no harm is caused by e-collars as the current is very small.

For the same reason that e-collars do not shock, they also cannot 'burn'. The 5 mJ pulse is nowhere near sufficient to deliver a burn. A Federal Court judgment in Australia found that e-collars are "incapable of inflicting burns upon dogs". [32] Yet, campaigners continue to recount anecdotes to suggest that they can.

Some anecdotes are often down to mistaken diagnosis. As with any harness or collar, an overly tight-fitting device can create pressure sores or an allergic reaction. But this is a question of appropriate use and not to do with shocks or burns.

c) The veterinary comparables

The vets who support e-collars believe that the welfare advantages are overwhelming. [33] They also acknowledge that their own profession regularly makes highly aversive interventions into the lives of animals. At the more extreme end, the spaying and castration of dogs, is increasingly questioned. [34] Other aversive interventions include the use of the cone-shaped "Elizabethan" collars and the dragging of dogs into veterinary practices against their will. [35] [36]

Other objections to e-collars

a) All aversive training is wrong

This is the creedal statement of the reward-only training movement. [37]

Being 'purely positive' is an attractive sounding doctrine to preach. The dog gets treats but never a consequence for an unwanted behaviour.

Such reward-based training does have its place. It is effective at reinforcing *desirable* behaviours like a dog coming to heel.

Yet the theory of operant conditioning indicates that it has no impact on stopping *unwanted* behaviours – like chasing sheep.

The reward-only dogma would ban any form of e-collar training, however low the pulse. It creates unfair pressure on dog owners who are struggling to prevent unwanted behaviours. A January 2024 survey found 97% of "pandemic puppy" owners were facing behavioural problems, with 82% using aversive methods such as shouting and yanking of collars. [38]

There are millions of dog and cat owners who are struggling to train and contain their pets in the EU.

Instead of making them feel guilty for predictably 'failing' with reward-only training, many issues could be simply and effectively solved, if they were guided towards a balanced training mix with well-evidenced methodologies for tackling unwanted, as well as wanted behaviours.

b) Reward-only training can stop predation

Supporters of reward-only training point to one research paper suggesting that it can reduce predatory attacks. This is the "China paper" - an MSc thesis by a student at a lowly-rated university in England. [39] Given that its claim clashes with the scientific consensus, the paper has been forensically unpicked by one of the world's top experts on animal psychology - Professor Douglas Elliffe from the University of Auckland.

Professor Elliffe wrote that, “as evidence to inform evidence-based policy making, it is very seriously flawed and should not be relied on”. [40] Campaigners have never sought to rebut this eminent professor’s critique. Nor have they responded to the revelation that the research China analysed, was conducted by academics who had previously urged politicians to ban e-collar training. [41]

c) Relying on leads obviates the need for e-collar training

Responsible dog owners should use leads whenever they think their dog could cause harm. The problem is that dogs can smell out-of-sight predatory targets long before their owners. And in the countryside, targets such as livestock, rabbits and birds are in every direction. Yet keeping a dog permanently on a lead is not approved by vets because of the psychological and physical need for off-lead exercise. [42]

So how can owners living in the countryside give their dogs freedom, without them racing off when they detect animals of interest? The only solution is effective training. And the only effective training for dogs with high prey drives is with e-collars.

d) Dogs are escaping so e-collar training is irrelevant

The dilemma around deliberately giving freedom to dogs, leads us to the problem of unplanned escapes. Powerful dogs easily break free from their owners’ grasp or escape from homes. UK police found that 66% of dogs who chased sheep had no owner nearby. [43] Campaigners argue that in these situations, e-collars cannot be effective because the owner is not present to administer a pulse.

This highlights a significant misunderstanding about how e-collars work. Their central purpose is not to be a virtual leash for when a dog is in the process of escaping from its owner. It is instead to create a lasting aversion response to the predatory target which works in all situations. After an escape, the e-collar trained dog considers any sheep it encounters as able to deliver a pulse, not its owner.

e) Dogs become confused by e-collar training

Here the narrative suggests that owners are unable to convey the desired message to their dog through an e-collar. This is a risk with any form of training. It speaks to the need for quality training rather than a problem with any given tool.

Furthermore, e-collars can enhance the owner’s ability to communicate with their dog. Through very low-level pulses, the owner can provide the dog with much more precise guidance than the ubiquitous alternatives of shouting or yanking a lead. A pulse can be delivered the moment a dog lunges towards its target, and stopped as soon as it turns away. This flexible communication can be used whether the dog is nearby or hundreds of yards away.

Options for the European Institutions

There is clearly an issue with the text on “electric current”. Its broad scope would ban therapeutic interventions by vets in boarding and breeding facilities using TENS devices.

The inclusion of e-collars in the scope creates a serious of unwanted consequences which would be highly detrimental to animal welfare. Many member states believe that their existing animal welfare legislation covers the theoretical risks of their abuse and misuse.

That suggests that the Commission could decide that any mention of electric current in the text is unwarranted and achieve this by the removal of Article 15.3 (c).

Alternatively, it could examine how e-collars are regulated in Australia. There, so long as product standards are adhered to, the user of an e-collar is deemed to have complied with animal welfare legislation. [44]

Or the EU could follow the example of New Zealand's Department of Conservation which is so convinced of the unique effectiveness of e-collars that it mandates their use in some national parks. [45]

Unless the unintended consequences are addressed the current text is likely to lead to considerable concerns in the incoming European Parliament.

Sourcing

[1] The text of the directive. Article 15 deals with "electric current": https://food.ec.europa.eu/system/files/2023-12/aw_reg-proposal_2023-769_dog-cat-trace.pdf
[2] Vets use electrotherapy post-operatively to help maintain muscle mass and to alleviate chronic conditions such as osteoarthritis. The devices are known as TENS machines. (Transcutaneous Electrical Nerve Stimulation / neurostimulation électrique transcutanée.) https://clinique-veterinaire-de-la-riviere.com/lelectrostimulation-pour-les-chiens/
[3] The UK Government halted attempts to pass a ban on remote e-collars which would have come into force in February 2024: https://statutoryinstruments.parliament.uk/instrument/vUkNBtTH/timeline/JTqCGOUW/
[4] Preventing attacks by wolves: https://wolfwatcher.org/wp-content/uploads/2015/01/Rossler-et-al-2012-shock-collars.pdf and by coyotes: https://www.researchgate.net/publication/258098937_Coyote_predation_on_domestic_sheep_deterred_with_electronic_dog-training_collar
[5] Geo-fenced e-collars are used in Australia and the USA to move livestock without physical fencing: https://am.gallagher.com/en-CA/new-products/eShepherd as well as by the RSPB https://community.rspb.org.uk/getinvolved/corncrake-calling/b/corncrake-calling/posts/nofence-collars---saviour-of-abandoned-crofts-for-corncrakes-and-other-wildlife
[6] The Guardian: e-collar avoidance training is mandatory in New Zealand as it is the "most effective tool" for stopping dogs attacking wildlife https://www.theguardian.com/environment/2022/jul/02/kiwi-cant-defend-themselves-new-zealand-wrestles-with-how-to-save-an-icon The New Zealand Government pays for e-collar training to stop dog attacks on native species: https://www.stuff.co.nz/national/120185885/whio-aversion-training-being-offered-to-owners-of-farm-hunting-dogs
[7] Australian vets recommend them to stop dogs attacking snakes as "nothing else nearly as effective" - see page 14: https://issuu.com/engagemedia/docs/vet_practice_nov_2017
Also 25% of working dogs used by Australian Government agencies were trained with e-collars: Australian Government survey: p33 https://elkana.info/downloads/working-dog-survey.pdf
[8] At least 30,000 livestock killed by wolves in the EU each year: https://rewildingeurope.com/blog/the-key-to-living-with-wolves-in-europe-ramping-up-livestock-protection-measures
[9]An academic study found that 3.3% of dog owners use them to them to train dogs, 0.9% as containment fence systems and 1.4% as anti-bark collars.: https://bmcvetres.biomedcentral.com/articles/10.1186/1746-6148-8-93
[10] EU has 72 million dogs this would indicate around 3.6 million dogs are being trained with e-collars across the member states: https://ec.europa.eu/commission/presscorner/detail/en/qanda_23_6253
[11] Andrew Lloyd Webber using an e-collar to protect his cats: https://www.thetimes.co.uk/article/lloyd-webber-in-hot-water-over-cat-collars-ftb5gbncj
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2830451/

[12] Peer-reviewed research on e-collar effectiveness:

- "All 1,156 dogs displayed avoidance… after the first training session", Dale; https://www.sciencedirect.com/science/article/abs/pii/S0168159117300746
- "The collar averted all 13 attempted attacks on lambs", Andelt: https://www.researchgate.net/publication/258098937_Coyote_predation_on_domestic_sheep_deterred_with_electronic_dog-training_collar
- e-collars "resulted in complete and permanent elimination of aggression in all of the 36 dogs tested", Tortora: https://cpb-us-w2.wpmucdn.com/about.illinoisstate.edu/dist/6/45/files/2019/10/tortora-1983-safety-signal-training-elimination-of-avoidance-motivated-aggression-in-dogs.pdf
- "no dogs showed interest in… sheep in the path test", Christiansen: https://www.ncbi.nlm.nih.gov/pubmed/11278032
- the aversion response "lasts at least 1 year after training", Dale and Statham: https://www.researchbank.ac.nz/handle/10652/2630
- e-collar training "shows stronger 'learning effect'", Salgirli: https://www.semanticscholar.org/paper/Comparison-of-learning-effects-and-stress-between-3-Salgirli-Schalke/6e2a2acbad0a3a9d1a630fa804c0e54536f17251
- "the most effective" training, see page 6, Howell and Bennett: https://www.sciencedirect.com/science/article/abs/pii/S016815912030071X

[13] "The closest relative to the domestic dog is undeniably the grey wolf, from which the dog differs by only 0.04% in nuclear coding-DNA sequence": https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2830451/

[14] The Telegraph reported that there were four times more dog attacks and three times more dogs being shot in Wales than in other parts of the country: https://www.telegraph.co.uk/news/2022/06/11/gareth-wyn-jones-joins-farmers-calls-reversal-electric-dog-collar/

[15] Former Welsh Secretary David Jones MP said Cardiff's ban on dog e-collars has "failed" and is "leading to many more animal deaths.": https://www.pressreader.com/uk/the-sunday-telegraph/20220327/281797107498632

[16] Plaid Cymru's Peredur Owen Griffiths MS: "E-collars act like a vaccine against a dog's desire to chase sheep… being shot by a farmer is clearly far worse for the dog than a one-off startle from an e-collar." https://www.farminguk.com/news/dog-attacks-on-sheep-due-to-failed-e-collar-training-ban_60113.html

[17] NFU Mutual insurance data shows a four-fold increase in attacks on dogs in three years:

NFU Mutual on dog attacks	**2020**	**2021**	**2022**	**2023**	**Percentage Increase 2020-2023**
UK Total	£1,300,000	£1,520,000	£1,800,000	£2,400,000	Up 85%
Wales	**£165,000**	£306,000	£439,000	**£883,000**	**Up 335%**
UK exc. Wales	**£1,135,000**	£1,214,000	£1,361,000	**£1,517,000**	**Up 34%**

- 2023 NFU Mutual data: https://nation.cymru/news/complacency-kills-as-cost-of-dog-attacks-on-livestock-in-wales-more-than-doubles/
- 2022 NFU Mutual data: https://www.nfumutual.co.uk/media-centre/uk-cost-of-dog-attacks-rises-by-50-causing-needless-suffering-of-sheep/
- 2021 NFU Mutual data: https://www.fwi.co.uk/news/crime/dog-attacks-on-livestock-cost-farmers-1-5m-in-2021
- 2020 NFU Mutual data: https://www.countytimes.co.uk/news/19057143.cost-dog-attacks-animals-concerns-remain/

[18] Ireland's legal definition on animal cruelty: https://www.irishstatutebook.ie/eli/2013/act/15/section/12/enacted/en/html#sec12
[19] 80 sheep killed in one attack: https://www.irishexaminer.com/farming/arid-41040467.html
[20] In Scotland no law enforcement agency has prosecuted or even issued a written complaint for e-collar misuse: Scottish Government Review 2021 https://www.gov.scot/publications/guidance-dog-training-aids-review/pages/5/
[21] In England the RSPCA has never found a case of e-collar abuse that merited prosecution despite receiving calls about animal welfare problems "every 30 seconds": https://www.rspca.org.uk/utilities/contactus/reportcrueltyhttps://www.rspca.org.uk/utilities/contactus/reportcruelty
[22] The Kennel Club told the Scottish Government that it had no evidence of "misuse or abuse" of e-collar training: https://consult.gov.scot/animal-welfare/electronic-training-aids/consultation/view_respondent?show_all_questions=0&sort=excerpt&order=ascending&_q__text=kennel+club&uuId=907963365[22] MEP referred to e-collars as being "torture devices": https://www.europarl.europa.eu/doceo/document/E-9-2022-001983_EN.html
[23] Peer-reviewed research on e-collar safety:

- "no negative effect on the dogs", Christiansen: https://www.ncbi.nlm.nih.gov/pubmed/11278032
- Dogs "did not show considerable or persistent stress indicators", Schalke: https://www.sciencedirect.com/science/article/abs/pii/S0168159106003820
- "induced less stress" than other training methods, Salgirli: https://www.semanticscholar.org/paper/Comparison-of-learning-effects-and-stress-between-3-Salgirli-Schalke/6e2a2acbad0a3a9d1a630fa804c0e54536f17251
- e-collar use led to "significant increases in the dogs' emotional stability", Tortora: https://cpb-us-w2.wpmucdn.com/about.illinoisstate.edu/dist/6/45/files/2019/10/tortora-1983-safety-signal-training-elimination-of-avoidance-motivated-aggression-in-dogs.pdf

[24] Companion Animal Welfare Councils says term "shock" is misleading: https://business.senedd.wales/documents/s18527/03.06.2013%20Correspondence%20-%20Companion%20Animal%20Welfare%20Council%20to%20the%20Chair.pdf
[25] Officials in London use the term "electronic collars: https://statutoryinstruments.parliament.uk/instrument/vUkNBtTH/timeline/JTqCGOUW/
[26] Officials in Edinburgh use the term: "e-collars": https://www.gov.scot/publications/report-use-handheld-remote-controlled-training-devices-e-collars-dog-training-scottish-animal-welfare-commission/pages/7/
[27] The international product standard of the Electronic Collar Manufacturers Association sets a a maximum output of 5 mJ - see page 7: http://ecma.eu.com/wp-content/uploads/2016/10/120411amc-ECMA-Technical-Requirement-6-0-FINAL-APPROVED.pdf
[28] Livestock fences, which any animal can touch, go up to 15,000 mJ, although the technology tries to limit the shock to 5,000 mJ: https://www.irishsales.ie/?product=ubison-15-000-mains-fencer-15joule
[29] MEP referred to e-collars as being "torture devices": https://www.europarl.europa.eu/doceo/document/E-9-2022-001983_EN.html
[30] Senator Regina Doherty said e-collars can deliver "6,000 volts" they were "incredibly cruel" and "barbaric": https://www.finegael.ie/as-a-nation-of-dog-lovers-we-must-ban-electronic-shock-collars-doherty/
[31] A Van de Graaff generator can safely create 200,000 volts: https://spark.iop.org/collections/van-de-graaff-generator and https://www.youtube.com/watch?v=ubZuSZYVBng
[32] Federal Court Judge, Australia found that e-collars are "incapable of inflicting burns upon dogs", paragraph 91: https://jade.io/article/332481
[33] Vets' letter in The Times supports e-collars: https://www.thetimes.co.uk/article/nhs-patients-and-broken-drugs-industry-smvxfqj2m
[34] Aversive veterinary interventions: spaying and castration: https://www.naiaonline.org/pdfs/LongTermHealthEffectsOfSpayNeuterInDogs.pdf

[35] Aversive veterinary interventions: Elizabethan collars
https://www.researchgate.net/publication/339424007_The_Cone_of_Shame_Welfare_Implications_of_Elizabethan_Collar_Use_on_Dogs_and_Cats_as_Reported_by_their_Owners
[36] Aversive veterinary interventions: forcing dogs into veterinary practices: https://www.sciencedirect.com/science/article/abs/pii/S109002330800169X
[37] The belief that reward only training is the most effective form of training dogs is widespread despite it not fitting established behavioural psychology or academic research: https://www.dogstrust.org.uk/dog-advice/training/techniques/positive-reinforcement-training-with-rewards
[38] RVC survey on "pandemic puppies" in the UK found that 97% were confronting at least one behavioural problem and 82% were using aversive methods to control their dogs: https://www.theguardian.com/lifeandstyle/2024/jan/22/pandemic-puppy-owners-still-struggling-with-their-dogs-behaviour-survey-shows
[39] Lucy China MSc, University of Lincoln, Efficacy of Dog Training With and Without Remote Electronic Collars vs. a Focus on Positive Reinforcement: https://www.frontiersin.org/articles/10.3389/fvets.2020.00508/full
[40] Professor Douglas Elliffe, Deputy Dean of the Faculty of Science of the University of Auckland, is one of the world's leading experts on animal psychology. He concluded that the China paper and its underlying research is "very seriously flawed": https://joinardo.com/wp-content/uploads/2020/12/D-201127-Lincoln-Review-Professor-Elliffe.pdf
[41] The academics who produced the underlying research in the China paper had previously campaigned for a ban on e-collars:
- Professor Daniel Mills had previously written to the UK Government urging a ban on e-collars. See p79: https://drive.google.com/file/d/15fnxt71d3OL7fQUPfzKriTWJCxXFGEaO/view?usp=share_link
- two other researchers, Nina Cracknell and Hannah Wright, had signed a petition calling for a ban. https://web.archive.org/web/20090821171442/http://www.gopetition.co.uk/online/27913/signatures-page14.html
[42] "89% of veterinary professionals agree that the welfare of dogs will suffer…if dogs are required to be kept on leads", Page 9: https://www.pdsa.org.uk/media/4371/paw-2018-full-web-ready.pdf
The Kennel Club also says: 'Without sufficient off-lead exercise, dogs will likely be bored, have excess energy and be at a greater risk of straying.': https://www.thekennelclub.org.uk/about-us/campaigns/livestock-worrying/
[43] UK Police date shows that in around 66 % of livestock worrying occurs when the owner/walker of the dog is not present, Page 12: https://npcc.police.uk/Publication/livestock%20worrying.pdf
The% Kennel Club accepts that escape from human control is a widespread problem and approaches 90% of livestock worrying cases in North Wales: https://www.thekennelclub.org.uk/media-centre/2019/march/kennel-club-launches-whereisyourdog-campaign-in-response-to-rising-livestock-attacks/
[44] In Australia e-collars product standards are incorporated into legislation in:
Western Australia - see page 38: https://www.agric.wa.gov.au/sites/gateway/files/Standards%20and%20Guidelines%20for%20the%20Health%20and%20Welfare%20of%20Dogs.pdf
Victoria - see sections 23 to 29: https://content.legislation.vic.gov.au/sites/default/files/9cc60c2d-63c4-35b6-95ad-c39254a5d170_19-133sra%20authorised.pdf
[45] New Zealand Government pays for e-collar training to stop dog attacks on native species: https://www.stuff.co.nz/national/120185885/whio-aversion-training-being-offered-to-owners-of-farm-hunting-dogs and https://www.doc.govt.nz/parks-and-recreation/know-before-you-go/dog-access/avian-awareness-and-avoidance-training/